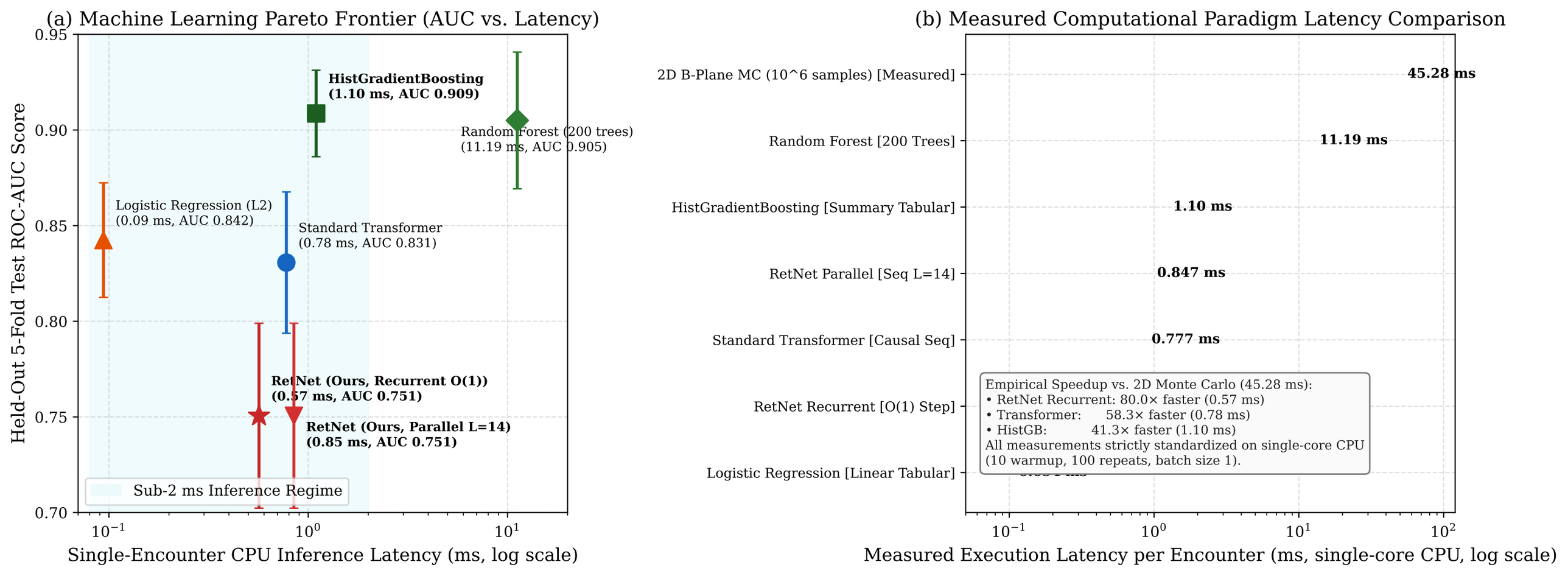
(a) Machine Learning Pareto Frontier (AUC vs. Latency)
Held-Out 5-Fold Test ROC-AUC Score
Single-Encounter CPU Inference Latency (ms, log scale)
HistGradientBoosting
(1.10 ms, AUC 0.909)
Random Forest (200 trees)
(11.19 ms, AUC 0.905)
Logistic Regression (L2)
(0.09 ms, AUC 0.842)
Standard Transformer
(0.78 ms, AUC 0.831)
RetNet (Ours, Recurrent O(1))
(0.57 ms, AUC 0.751)
RetNet (Ours, Parallel L=14)
(0.85 ms, AUC 0.751)
Sub-2 ms Inference Regime
(b) Measured Computational Paradigm Latency Comparison
2D B-Plane MC (10^6 samples) [Measured]
Random Forest [200 Trees]
HistGradientBoosting [Summary Tabular]
RetNet Parallel [Seq L=14]
Standard Transformer [Causal Seq]
RetNet Recurrent [O(1) Step]
Logistic Regression [Linear Tabular]
45.28 ms
11.19 ms
1.10 ms
0.847 ms
0.777 ms
Empirical Speedup vs. 2D Monte Carlo (45.28 ms):
• RetNet Recurrent: 80.0× faster (0.57 ms)
• Transformer: 58.3× faster (0.78 ms)
• HistGB: 41.3× faster (1.10 ms)
All measurements strictly standardized on single-core CPU
(10 warmup, 100 repeats, batch size 1).
Measured Execution Latency per Encounter (ms, single-core CPU, log scale)